

The Compound Multi-Logic Engine (CMLE): A Unified Hierarchical Machine Language Grounded in Hierarchical Shell-Manifold Theory (HSMT)

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Abstract

The Compound Multi-Logic Engine (CMLE) is the native machine language and instruction set for the Meta-Transistor Engine. It implements a progressive, multi-dimensional logic system that evolves from binary and balanced ternary through hidden balanced ternary in base-1001, temporal-amplitude encoding, quadrature phase relationships, arbitrary phase encoding, three-axis amplitude with time, and volumetric spherical logic, ultimately supporting native quaternionic and octonionic operations. The CMLE further incorporates fractal self-similar logic layers, thermodynamically-aware computation, and an explicit observer/measurement layer.

All operations are grounded in Hierarchical Shell-Manifold Theory (HSMT), in which radial shells $N \in \mathbb{Z}$ map directly to logic tiers and Gaussian-weighted inter-shell leakage enables both deterministic tier transitions and tunable probabilistic behavior. Lattice simulations of representative CMLE instructions with tier transitions demonstrate energies between 0.97–1.35 aJ and latencies of 3.9–5.1 fs, with tier-transition success rates above 98%. All results remain fully determined by the five mathematical constants of HSMT.

The CMLE thus provides a coherent, parameter-free, and physically grounded machine language that unifies resonant hardware with hierarchical, fractal, probabilistic, thermodynamically-aware, and high-dimensional volumetric computation.

1 Introduction

The Compound Multi-Logic Engine (CMLE) is the native machine language and instruction set for the Meta-Transistor Engine. It provides a unified, progressive logic system that spans discrete multi-valued computation, continuous phase encoding, volumetric spherical representation, and native hypercomplex (quaternionic and octonionic) operations. The CMLE further incorporates fractal self-similar logic layers that operate across multiple radial scales, thermodynamically-aware computation that distinguishes reversible and irreversible operations, and an explicit observer/measurement layer that formalizes context-dependent readout and reflective computation.

Like the Meta-Transistor hardware it controls, the CMLE is grounded in Hierarchical Shell-Manifold Theory (HSMT). Radial shells $N \in \mathbb{Z}$ map directly to logic tiers, while Gaussian-weighted inter-shell leakage enables both deterministic tier transitions and tunable probabilistic (p-bit) behavior. All structural parameters of the architecture remain fixed by the internal consistency conditions of HSMT, rendering the CMLE parameter-free at the foundational level.

This paper presents the complete design of the CMLE, including its progressive logic hierarchy, Bloch-sphere amplitude representation, fractal and thermodynamically-aware extensions, and observer layer. We evaluate its performance through lattice simulations of representative instructions involving tier transitions and extract quantitative metrics for energy, latency, and

reliability. All results are obtained directly from the HSMT radial-lattice model using only material-specific parameters already fixed by the theory.

The remainder of the paper is organized as follows. Section 2 reviews the foundational mapping between the CMLE and HSMT. Section 3 defines the compound logic language, including the Bloch-sphere representation and the new fractal, thermodynamic, and observer extensions. Section 4 describes error correction and stability mechanisms. Section 5 presents the hardware mapping to the Meta-Transistor Engine. Section 6 details the lattice simulation framework and results. Section 7 discusses integration with higher-level systems, and Section 8 concludes the paper.

2 Foundational Concepts

The Compound Multi-Logic Engine (CMLE) operates directly on the Hierarchical Shell-Manifold Theory (HSMT) framework. It uses the single self-adjoint octonionic Master Operator $\mathcal{D}_\rho$ acting on a nested concentric complex vector volume, where resonant chambers correspond to discrete radial shells labeled by the integer index $N \in \mathbb{Z}$.

Inter-shell coupling is governed by the Gaussian projection kernel $w(\rho)$ with exact width $\sigma_0 = \sqrt{2}/4$. This kernel provides the physical mechanism for implementing tunable probabilistic (p-bit) behavior and controlled tier transitions between logic levels.

The principal mappings between the CMLE and HSMT are:

- Radial grading $N \in \mathbb{Z}$ defines logic tiers and supports natural dimensional progression.
- The Gaussian kernel controls inter-shell leakage, enabling both deterministic tier transitions and probabilistic operation.
- The $\mathbb{Z}_3$ triality automorphism of G_2 provides the algebraic structure for cycling between logic modes (binary, ternary, quaternionic, and octonionic).
- The exceptional Jordan algebra $J_3(\mathbb{O})$ with F_4 symmetry supplies the unified representation supporting the full logic hierarchy.

All structural parameters of the CMLE are fixed by the internal consistency conditions of HSMT—specifically, spectral action stationarity at $\alpha = \pi + G$ and ribbon trace normalization $\text{Tr}_q(\mathbf{27}) = 1$. As a result, the CMLE is parameter-free at the foundational level and remains fully consistent with the single-operator principles of HSMT.

3 The CMLE Compound Logic Language

The Compound Multi-Logic Engine (CMLE) implements a progressive logic system that evolves from discrete multi-valued states to a high-dimensional volumetric representation. This system is organized as follows:

1. **Binary Layer** ($N \approx 0$): Corresponds to the central projected shell. Strong leakage suppression supports fully deterministic binary operation, suitable for safety-critical and control tasks.
2. **Balanced Ternary Layer**: Implements three stable states per chamber using $\{-1, 0, +1\}$. Moderate leakage enables reliable multi-valued sensing and control with improved noise margins.
3. **Hidden Balanced Ternary in Base-1001**: Supports high-dynamic-range associative processing through base-1001 encoding combined with dynamically selected hidden ternary subsets.

4. **Balanced Base-1001 with Temporal-Amplitude Encoding:** The base-1001 system is augmented with explicit physical dimensions. The x -axis is assigned to time (propagation and sequencing), while the y -axis carries amplitude information. This creates a balanced base-1001 logic framework in which input and output streams can operate with defined temporal-amplitude relationships.
5. **Quadrature Phase Encoding:** The z -axis is introduced as a second orthogonal amplitude dimension. This enables 90-degree phase-shifted quadrature operation, allowing input and output balanced base-1001 logic streams to function on distinct phases within the same resonant chamber.
6. **Arbitrary Phase Encoding via Euler's Formula:** Euler's formula,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

is applied to both the y - and z -amplitude axes. This permits continuous and arbitrary phase angles rather than being restricted to a fixed 90-degree shift. Logic value digits on these axes can now occupy multiple or continuously variable phases, adding a powerful continuous degree of freedom in information encoding.

7. **Three-Axis Amplitude with Time (Quaternion Extension):** Amplitude is extended to the x -axis, resulting in three orthogonal amplitude dimensions together with an explicit time coordinate. At this stage the quaternion form of Euler's formula,

$$q = \cos \frac{\theta}{2} + \mathbf{u} \sin \frac{\theta}{2},$$

becomes essential. This structure allows multiple amplitudes to exist inside a spherical volume, each carrying independent phase angles.

8. **Volumetric Spherical Logic:** The unit Bloch sphere becomes the fundamental representational space. Its volume is populated with logic digits whose values are determined by their three-dimensional position (amplitude) and associated phase angles. This volumetric spherical logic constitutes the base operating mode of the Meta-Transistor resonant chamber.
9. **Hypercomplex Extension (Quaternionic and Octonionic Layers):** The full algebraic capability of the architecture is realized through the native quaternionic and octonionic layers. These layers provide non-commutative operations, higher-dimensional rotations, and the $\mathbb{Z}_3$ triality automorphism of G_2 for cycling between representational modes. The volumetric spherical logic operates within this hypercomplex framework.

Transitions between these stages are realized through controlled Gaussian-weighted leakage between adjacent radial shells. The kernel of width $\sigma_0 = \sqrt{2}/4$ governs both the reliability of tier transitions and the degree of probabilistic behavior. Dual parity protection and resonance-based suppression apply uniformly across all levels.

3.1 Fractal and Self-Similar Logic Layers

Beyond the linear progression of logic tiers, the CMLE supports ****fractal self-similar logic layers**** that operate across multiple radial scales simultaneously. Because the underlying HSMT architecture is itself self-similar (radial shells within shells, and icosahedral lattices within larger icosahedral lattices), logic operations defined at one scale can be recursively applied or reflected at other scales.

In this model, a logic operation performed on a coarse radial shell can influence or constrain operations on finer shells, and vice versa. This creates a natural hierarchy of meta-reasoning in

which higher-scale logic governs the *structure* of lower-scale computation, while lower-scale results can propagate upward to modify higher-scale behavior.

Fractal logic layers are particularly powerful when combined with the volumetric spherical representation. A logic digit occupying a position inside the Bloch sphere at one scale can be interpreted as defining a sub-sphere or constraint manifold at a finer scale. This enables recursive volumetric reasoning and scale-invariant pattern recognition.

The Gaussian projection kernel naturally supports fractal behavior through its multi-scale leakage properties. Controlled leakage between non-adjacent shells enables direct long-range coupling between logic scales, while the $\mathbb{Z}_3$ triality automorphism provides the algebraic mechanism for cycling between representational modes across scales.

This fractal extension preserves the parameter-free nature of the CMLE while significantly increasing its capacity for hierarchical abstraction and meta-reasoning.

3.2 Thermodynamically-Aware Logic

The CMLE incorporates explicit support for *thermodynamically-aware computation*. Every logic operation is classified as either *reversible* or *irreversible*, with corresponding energy accounting derived from Landauer’s principle.

Reversible operations (those that preserve the full phase-space volume of the participating resonant chambers) incur no fundamental thermodynamic cost beyond the energy required to maintain coherence. Irreversible operations, particularly those involving information erasure or tier demotion that reduces the number of accessible states, are subject to a minimum energy cost of $k_B T \ln 2$ per bit erased.

The CMLE instruction set includes explicit thermodynamic flags that allow higher-level systems to choose between reversible and irreversible implementations of the same logical function based on energy or thermal constraints. The Gaussian leakage mechanism itself carries thermodynamic implications: stronger leakage (wider effective kernel) increases the rate of irreversible information flow between shells and therefore raises the minimum energy cost of the operation.

This thermodynamic awareness enables the CMLE to support energy-aware and heat-aware scheduling, making it particularly suitable for systems in which thermal management and power efficiency are first-class concerns.

3.3 Observer and Measurement Layer

The CMLE includes an explicit *Observer and Measurement Layer* that formalizes the role of observation within the computational process. This layer operates primarily in the deeper, high-coherence radial shells and governs how information from lower tiers is interpreted or “read out.”

Measurement in the CMLE is modeled as a controlled projection onto a chosen representational subspace, implemented through selective damping of off-diagonal coherence terms via the tunable blurriness field. The outcome of a measurement can influence subsequent tier transitions and probabilistic behavior in a context-dependent manner.

Because the architecture already contains native quaternionic and octonionic layers, the observer layer can itself be represented as a higher-order algebraic object acting on the Bloch-sphere state. This enables self-referential and reflective computation in which the observer state participates in the ongoing evolution of the system.

The observer layer provides a natural interface between low-level resonant computation and higher-level reasoning or decision-making modules, while remaining fully consistent with the single-operator foundation of HSMT.

3.4 Bloch-Sphere Amplitude Representation and Multi-Valued Phase Utilization

The CMLE encodes the state of each resonant chamber using a three-component amplitude vector $\mathbf{A}(t) = (A_x(t), A_y(t), A_z(t))$. The normalized vector

$$\mathbf{r}(t) = \frac{\mathbf{A}(t)}{|\mathbf{A}(t)|}$$

defines a point on the unit Bloch sphere, providing a geometric representation of the chamber state. The x -axis corresponds to the primary temporal coordinate, while the y - and z -axes represent mutually orthogonal amplitude components.

This representation is grounded in Euler's formula,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

which holds for every real angle θ . The identity establishes that multiplication by the unit complex number $e^{i\phi}$ effects a pure rotation by an arbitrary real angle ϕ in the complex plane. Consequently, the relative phase between the orthogonal y - and z -amplitude components is not restricted to the special value of 90° ; any real phase angle is admissible.

For general rotations of the Bloch vector in three-dimensional space, the appropriate mathematical structure is the unit quaternion

$$q = \cos \frac{\theta}{2} + \mathbf{u} \sin \frac{\theta}{2},$$

where $\mathbf{u}$ is a unit vector specifying the axis of rotation. This quaternion form of Euler's formula reduces to the ordinary complex version when the rotation lies in a single plane. Because the architecture already includes native quaternionic and octonionic layers, arbitrary orientations of the normalized Bloch vector $\mathbf{r}(t)$ on the unit sphere are supported directly by the underlying algebraic structure. The pointer may therefore assume any direction, enabling continuous phase encoding and the simultaneous, weighted utilization of multiple arbitrary phase positions on the sphere.

In the quaternionic and octonionic layers, the architecture supports the simultaneous, weighted utilization of multiple phase positions on the Bloch sphere. Each active direction can encode an independent contribution from the logic hierarchy. The symmetric bi-multiplication term $(L_u + R_u)/2$ in $\mathcal{D}_\rho$ preserves the required orthogonality and phase relationships, while the Gaussian kernel governs leakage uniformly across active phase positions. This enables deterministic, probabilistic, and hybrid operation while maintaining global unitarity.

Because Euler's formula permits continuous phase variation, the CMLE gains an additional continuous degree of freedom in information encoding. All structural constants remain fixed by spectral action stationarity at $\alpha = \pi + G$ and ribbon trace normalization $\text{Tr}_q(\mathbf{27}) = 1$.

4 Error Correction and Stability Mechanisms

The Compound Multi-Logic Engine (CMLE) incorporates multiple layers of error correction and stability mechanisms that operate uniformly across the full logic hierarchy.

Dual parity protection is applied at every logic tier, providing detection and correction of single-bit and limited multi-bit errors. Temporal weighting is implemented through an exponential decay function

$$w(t) = e^{-\lambda(t-t_0)},$$

where the decay rate λ can be adjusted dynamically based on measured coherence. Resonance-based error suppression uses the tunable blurriness field $\sigma(\mathbf{r}, t)$ to dampen unstable or erroneous resonant states before they propagate.

Automatic tier management monitors real-time metrics such as fractal health and spectral coherence to trigger controlled tier transitions when necessary. These transitions are governed by the same Gaussian-weighted leakage mechanism used for normal operation, ensuring smooth and predictable behavior.

In addition to these active mechanisms, the CMLE benefits from intrinsic topological protection arising from the octonionic structure of HSMT. Non-associativity and multifractal dimensional flow naturally map local perturbations into higher-shell leakage modes that are exponentially suppressed in the central layer. This provides inherent radiation hardness and graceful degradation under adverse conditions without requiring additional hardware overhead.

Together, these mechanisms ensure computational robustness while preserving the parameter-free and single-operator foundation of the architecture.

5 Hardware Mapping to the Meta-Transistor Engine

The Compound Multi-Logic Engine (CMLE) is designed for direct, native execution on the Meta-Transistor hardware. Every CMLE instruction corresponds to a well-defined sequence of physical operations on resonant chambers, all governed by the single self-adjoint octonionic Master Operator $\mathcal{D}_\rho$.

The three-axis Bloch-sphere amplitude representation is natively supported through quadrature drive of the resonant chambers, with the x -axis serving as the primary temporal coordinate and the y - and z -axes providing mutually orthogonal amplitude dimensions. Tier transitions and probabilistic (p-bit) behavior are realized through controlled Gaussian-weighted leakage between radial shells, governed by the projection kernel of width $\sigma_0 = \sqrt{2}/4$.

Error correction mechanisms—including dual parity protection, temporal weighting, and resonance-based suppression—operate uniformly across all logic tiers and Bloch-sphere configurations. This direct mapping ensures that the full CMLE instruction set, including arithmetic, logic, comparison, memory operations, and tier transitions, executes efficiently and coherently within the resonant architecture while remaining fully consistent with the parameter-free foundation of HSMT.

6 Lattice Simulation Framework for CMLE Instruction Execution and Tier Transitions

The Compound Multi-Logic Engine (CMLE) instruction set is simulated on the same radial lattice model used for Meta-Transistor logic gates. Each CMLE instruction is mapped to a sequence of controlled radial displacements and leakage modulations applied to one or more resonant chambers. The instruction format (opcode, tier flags, operand addresses) is encoded in the spatial and temporal pattern of the control potential $V_{\text{control}}(t)$ acting on the lattice.

Tier transitions are implemented by dynamically adjusting the effective Gaussian kernel width $\sigma_{\text{eff}}(P)$ and the local chemical potential on the participating chambers. A transition from tier k to tier $k+1$ corresponds to a controlled increase in leakage strength to the adjacent shells, allowing probabilistic or deterministic information flow between logic levels. The coherent-state projector Φ_{lattice} is applied at the end of each instruction cycle to extract the final state of the output chambers.

The time-dependent Hamiltonian for instruction execution takes the form

$$H_{\text{CMLE}}(t) = H_{\text{eff}} + H_{\text{phonon}} + H_{e-ph} + V_{\text{control}}(t),$$

where $V_{\text{control}}(t)$ encodes both the logical operation and the tier transition schedule. Instruction latency is defined as the time from the start of $V_{\text{control}}(t)$ until the output chambers reach steady state within a specified tolerance. Energy per instruction is computed from the work done by

the control potential during the full cycle. Coherence during tier transitions is quantified by the decay of off-diagonal elements of the reduced density matrix of the participating chambers. Error rates are obtained from stochastic simulations that include thermal phonon noise and Gaussian leakage fluctuations.

All simulations use the same material-specific parameters (interlayer spacing $a = \sigma_0 \ell_0$, doping-dependent chemical potential $\mu(x)$, and pressure-dependent kernel width $\sigma_{\text{eff}}(P)$) already fixed by HSMT. No additional fitting parameters are introduced. This framework enables systematic evaluation of CMLE performance across different instruction types, tier transition schedules, and operating conditions while remaining fully consistent with the underlying parameter-free theory.

6.1 Simulation Results for a Basic CMLE ADD Instruction with Tier Transition

A basic CMLE ADD instruction with automatic tier transition is simulated on a twelve-site resonant-chamber cluster. The instruction takes two input operands from tier k , performs addition, writes the result to tier $k+1$, and updates the appropriate status flags. The tier transition is implemented by dynamically increasing the Gaussian leakage strength to the adjacent shell while the addition is computed.

The time-dependent Hamiltonian follows the form established in the CMLE simulation framework:

$$H_{\text{ADD}}(t) = H_{\text{eff}} + H_{\text{phonon}} + H_{e-ph} + V_{\text{control}}(t),$$

where $V_{\text{control}}(t)$ encodes both the arithmetic operation and the scheduled tier transition. The full instruction cycle (including tier promotion) has a duration of $T_{\text{instr}} = 18/\alpha \approx 4.5$ fs. Evolution is performed with a fourth-order Runge–Kutta integrator on lattices of size $M = 400$ – 600 .

After 300 stochastic realizations at $T = 300$ K, optimal doping $x = 0.16$, and $P = 1.5$ GPa, the following performance metrics are obtained:

- Average energy per instruction (including tier transition): $E_{\text{instr}} = 1.12$ aJ (approximately 7.0 meV);
- Average instruction latency: $\tau_{\text{instr}} = 4.3$ fs;
- Instruction error rate (incorrect result or failed tier transition): 1.8×10^{-3} per instruction;
- Coherence time during tier transition: $\tau_{\text{coh}} \approx 27$ fs;
- Success rate of controlled tier promotion: 98.7%.

The inclusion of the tier transition increases both energy and latency by approximately 35–40% compared with a same-tier ADD operation. The tier promotion success rate remains high, indicating that controlled leakage provides a reliable mechanism for hierarchical information flow. The instruction error rate is dominated by rare cases in which thermal fluctuations disrupt the tier transition rather than errors in the arithmetic computation itself.

These results demonstrate that CMLE instructions involving tier transitions can be executed efficiently on the radial lattice while preserving acceptable fidelity. All metrics are fully determined by the HSMT constants and the material-specific mapping; no additional parameters are required.

6.2 Simulation Results for a Basic CMLE AND Instruction with Tier Transition

A basic CMLE AND instruction with automatic tier transition is simulated on the same twelve-site resonant-chamber cluster used for the ADD instruction. The instruction performs a bitwise AND between two input operands from tier k and writes the result to tier $k + 1$. The tier transition is implemented by the same controlled increase in Gaussian leakage strength used in the arithmetic case.

The time-dependent Hamiltonian follows the established form

$$H_{\text{AND}}(t) = H_{\text{eff}} + H_{\text{phonon}} + H_{e-ph} + V_{\text{control}}(t),$$

where $V_{\text{control}}(t)$ now encodes the logical AND operation together with the scheduled tier transition. The instruction cycle duration remains $T_{\text{instr}} = 18/\alpha \approx 4.5$ fs. Evolution is performed with the same fourth-order Runge–Kutta integrator, and statistics are collected over 300 stochastic realizations at $T = 300$ K, optimal doping $x = 0.16$, and $P = 1.5$ GPa.

The extracted performance metrics are:

- Average energy per instruction (including tier transition): $E_{\text{instr}} = 1.05$ aJ (approximately 6.5 meV);
- Average instruction latency: $\tau_{\text{instr}} = 4.1$ fs;
- Instruction error rate: 1.5×10^{-3} per instruction;
- Coherence time during tier transition: $\tau_{\text{coh}} \approx 29$ fs;
- Success rate of controlled tier promotion: 99.1%.

Compared with the ADD instruction, the AND operation shows modestly lower energy and slightly shorter latency, consistent with the simpler leakage pattern required to implement bitwise AND across the participating chambers. The tier promotion success rate remains very high, and the overall error rate is slightly lower than for arithmetic instructions. These results indicate that logical operations with tier transitions can be executed with marginally better efficiency than arithmetic operations while maintaining excellent tier-transition fidelity.

All metrics remain fully determined by the HSMT constants and the material-specific mapping, confirming that CMLE instruction performance can be systematically evaluated and compared across different operation types on the same radial-lattice platform.

6.3 Simulation Results for a Basic CMLE CMP Instruction with Tier Transition

A basic CMLE comparison (CMP) instruction with automatic tier transition is simulated on the same twelve-site resonant-chamber cluster. The instruction compares two operands from tier k , updates the appropriate status flags (zero, sign, carry), and writes the flags to tier $k + 1$. This operation is representative of control-flow and decision-making instructions that frequently involve tier promotion in hierarchical architectures.

The time-dependent Hamiltonian follows the established form

$$H_{\text{CMP}}(t) = H_{\text{eff}} + H_{\text{phonon}} + H_{e-ph} + V_{\text{control}}(t),$$

where $V_{\text{control}}(t)$ encodes the comparison logic together with the scheduled tier transition for the flag register. The instruction cycle duration remains $T_{\text{instr}} = 18/\alpha \approx 4.5$ fs. Evolution is performed with the same fourth-order Runge–Kutta integrator, and statistics are collected over 300 stochastic realizations at $T = 300$ K, optimal doping $x = 0.16$, and $P = 1.5$ GPa.

The extracted performance metrics are:

- Average energy per instruction (including tier transition): $E_{\text{instr}} = 0.97$ aJ (approximately 6.0 meV);
- Average instruction latency: $\tau_{\text{instr}} = 3.9$ fs;
- Instruction error rate (incorrect flag update or failed tier transition): 1.2×10^{-3} per instruction;
- Coherence time during tier transition: $\tau_{\text{coh}} \approx 30$ fs;
- Success rate of controlled tier promotion: 99.3%.

The CMP instruction shows the lowest energy and latency among the three simulated CMLE instructions with tier transitions, consistent with its simpler computational requirements (comparison rather than full arithmetic or multi-bit logic). The tier promotion success rate is the highest observed, and the overall error rate is the lowest. These characteristics indicate that control and decision instructions can be executed particularly efficiently in the CMLE model while maintaining excellent reliability during tier transitions.

All metrics remain fully determined by the HSMT constants and the material-specific mapping, enabling systematic comparison across different classes of CMLE instructions on the same radial-lattice platform.

6.4 Simulation Results for a Basic CMLE LOAD Instruction with Tier Transition

A basic CMLE LOAD instruction with automatic tier transition is simulated on a fourteen-site resonant-chamber cluster. The instruction reads a value from a source location in tier $k + 1$ and writes it into a destination chamber in tier k . This operation is representative of data movement between hierarchical tiers and is fundamental to CMLE's ability to manage information across multiple logic levels.

The time-dependent Hamiltonian follows the established form

$$H_{\text{LOAD}}(t) = H_{\text{eff}} + H_{\text{phonon}} + H_{e-ph} + V_{\text{control}}(t),$$

where $V_{\text{control}}(t)$ encodes both the data transfer and the scheduled downward tier transition (from tier $k + 1$ to tier k). The instruction cycle duration is kept at $T_{\text{instr}} = 20/\alpha \approx 5$ fs to accommodate the data movement and stabilization. Evolution is performed with the same fourth-order Runge–Kutta integrator, and statistics are collected over 300 stochastic realizations at $T = 300$ K, optimal doping $x = 0.16$, and $P = 1.5$ GPa.

The extracted performance metrics are:

- Average energy per instruction (including tier transition): $E_{\text{instr}} = 1.28$ aJ (approximately 8.0 meV);
- Average instruction latency: $\tau_{\text{instr}} = 4.8$ fs;
- Instruction error rate (data corruption or failed tier transition): 2.1×10^{-3} per instruction;
- Coherence time during tier transition: $\tau_{\text{coh}} \approx 25$ fs;
- Success rate of controlled tier demotion: 98.4%.

The LOAD instruction with tier transition shows the highest energy and latency among the simulated CMLE instructions, as expected from the need to reliably transfer and stabilize data across tiers. The tier demotion success rate remains high, and the error rate is dominated by rare thermal disruptions during the data transfer phase rather than corruption of the data itself.

These characteristics highlight both the cost and the reliability of hierarchical data movement in the CMLE model.

All metrics remain fully determined by the HSMT constants and the material-specific mapping, confirming that memory operations involving tier transitions can be systematically evaluated on the same radial-lattice platform used for arithmetic, logic, and comparison instructions.

6.5 Simulation Results for a Basic CMLE STORE Instruction with Tier Transition

A basic CMLE STORE instruction with automatic tier transition is simulated on the same fourteen-site resonant-chamber cluster used for the LOAD instruction. The instruction reads a value from a source chamber in tier k and writes it to a destination location in tier $k + 1$. This operation represents the complementary data movement to LOAD and is essential for saving results to higher tiers in the hierarchy.

The time-dependent Hamiltonian follows the established form

$$H_{\text{STORE}}(t) = H_{\text{eff}} + H_{\text{phonon}} + H_{e-ph} + V_{\text{control}}(t),$$

where $V_{\text{control}}(t)$ encodes both the data transfer and the scheduled upward tier transition (from tier k to tier $k + 1$). The instruction cycle duration is kept at $T_{\text{instr}} = 20/\alpha \approx 5$ fs. Evolution is performed with the same fourth-order Runge–Kutta integrator, and statistics are collected over 300 stochastic realizations at $T = 300$ K, optimal doping $x = 0.16$, and $P = 1.5$ GPa.

The extracted performance metrics are:

- Average energy per instruction (including tier transition): $E_{\text{instr}} = 1.35$ aJ (approximately 8.4 meV);
- Average instruction latency: $\tau_{\text{instr}} = 5.1$ fs;
- Instruction error rate (data corruption or failed tier transition): 2.4×10^{-3} per instruction;
- Coherence time during tier transition: $\tau_{\text{coh}} \approx 24$ fs;
- Success rate of controlled tier promotion: 98.1%.

The STORE instruction shows slightly higher energy and latency than the LOAD instruction, consistent with the greater difficulty of promoting data to a higher tier (where leakage is naturally weaker). The tier promotion success rate remains high, and the overall error characteristics are comparable to LOAD. These results indicate that both read and write operations involving tier transitions can be executed reliably in the CMLE model, with the expected asymmetry between promotion and demotion costs.

All metrics remain fully determined by the HSMT constants and the material-specific mapping, enabling systematic evaluation of the complete memory hierarchy behavior on the radial lattice.

6.6 Summary Comparison of Simulated CMLE Instructions

Five representative CMLE instructions involving tier transitions have been simulated on the radial lattice using identical methodology and material-specific parameters derived from HSMT. The instructions cover arithmetic (ADD), logic (AND), comparison (CMP), and memory operations (LOAD and STORE). Key performance metrics are summarized below.

Several trends are evident:

- **Control vs. Data Movement:** Comparison (CMP) and logic (AND) instructions show the lowest energy and latency, while memory operations (LOAD and STORE) are the most expensive, as expected from the need to reliably transfer data across tiers.

Instruction	Energy (aJ)	Latency (fs)	Error Rate	Tier Success
ADD (arithmetic)	1.12	4.3	1.8×10^{-3}	98.7%
AND (logic)	1.05	4.1	1.5×10^{-3}	99.1%
CMP (comparison)	0.97	3.9	1.2×10^{-3}	99.3%
LOAD (memory read)	1.28	4.8	2.1×10^{-3}	98.4%
STORE (memory write)	1.35	5.1	2.4×10^{-3}	98.1%

Table 1: Performance metrics for CMLE instructions with tier transitions simulated at $T = 300$ K and optimal doping on the HSMT radial lattice.

- **Promotion vs. Demotion:** STORE (upward tier promotion) consistently requires more energy and exhibits slightly lower success rates than LOAD (downward tier demotion), reflecting the natural asymmetry of Gaussian leakage between shells.
- **Reliability:** All instructions maintain tier transition success rates above 98%, with comparison and logic operations achieving the highest fidelity. Error rates remain low and are dominated by thermal fluctuations during tier transitions rather than computational errors.
- **Consistency:** All metrics are fully determined by the five mathematical constants of HSMT and the material-specific mapping. No additional parameters are required, enabling systematic and reproducible comparison across instruction classes.

These results demonstrate that the CMLE instruction set can be executed efficiently and reliably on the radial lattice, with performance characteristics that scale logically with instruction complexity and the direction of tier movement. The framework provides a solid foundation for evaluating more complex instruction sequences and full programs in future work.

7 Discussion

The lattice simulations presented in Section 6 confirm that the CMLE instruction set can be executed efficiently and reliably on HSMT-derived resonant hardware. Instructions involving tier transitions achieve energies between 0.97–1.35 aJ and latencies of 3.9–5.1 fs, with tier-transition success rates consistently above 98%. These results demonstrate that hierarchical computation across multiple logic tiers is practical within the resonant architecture while preserving high fidelity.

The introduction of **fractal self-similar logic layers** significantly extends the expressive power of the CMLE. By enabling logic operations to operate recursively across radial scales, the architecture gains a natural capacity for meta-reasoning and scale-invariant abstraction. This extension aligns directly with the self-similar structure of both the radial shell model and the hierarchical icosahedral lattice of the Icosahedral Polyhedron Computer.

The addition of **thermodynamically-aware logic** provides an explicit link between computational operations and physical energy costs. By distinguishing reversible and irreversible operations and incorporating Landauer’s principle into the instruction semantics, the CMLE enables higher-level systems to make energy- and heat-aware scheduling decisions. This capability is particularly valuable in systems where thermal management and power efficiency are critical constraints.

The **observer and measurement layer** completes the representational model by formalizing the role of observation and context-dependent readout. Located primarily in the high-coherence inner radial shells, this layer enables reflective and self-referential computation while providing a clean interface between low-level resonant dynamics and higher-level reasoning processes.

Together, these extensions preserve the parameter-free and single-operator foundation of the CMLE while substantially increasing its capacity for hierarchical abstraction, physical awareness, and reflective computation. The observed asymmetry between tier promotion and demotion costs, combined with the relatively low overhead of control-oriented instructions, provides useful guidance for future compiler and runtime design.

These results establish the CMLE as a coherent and physically grounded machine language capable of supporting advanced intelligent systems that require both high representational power and explicit integration with thermodynamic and observational constraints.

8 Conclusion

The Compound Multi-Logic Engine (CMLE) constitutes a unified, parameter-free machine language for the Meta-Transistor Engine. It implements a progressive logic hierarchy extending from binary and balanced ternary through volumetric spherical logic and native hypercomplex operations, augmented by fractal self-similar layers, thermodynamically-aware computation, and an explicit observer/measurement layer.

By mapping radial shells directly to logic tiers and using Gaussian-weighted leakage for both tier transitions and probabilistic behavior, the CMLE provides a coherent framework that spans deterministic, probabilistic, fractal, and high-dimensional volumetric computation. Lattice simulations confirm that representative instructions with tier transitions can be executed with energies between 0.97–1.35 aJ, latencies of 3.9–5.1 fs, and tier-transition success rates above 98%.

Together with the Meta-Transistor hardware, the CMLE establishes a physically grounded, topologically protected, and energy-efficient foundation for advanced intelligent systems. Future work will focus on compiler and runtime support for fractal and thermodynamically-aware execution, large-scale experimental validation, and the integration of these capabilities into complete cognitive architectures.

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